

EDUCATION

UNIVERSITY OF CAMBRIDGE

PH.D. ARTIFICIAL INTELLIGENCE FOR
THE STUDY OF ENVIRONMENTAL RISKS
M.RES. ENVIRONMENTAL DATA SCIENCE

Oct. 2022 - Sep. 2026 | Cambridge, UK

UKRI EPSRC CDT Studentship (\$290k)

Dissertation: Understanding Regional Dynamic
Physical Risks: Advanced Geospatiotemporal
Modelling of Exposure & Vulnerability

STANFORD UNIVERSITY

M.A. PUBLIC POLICY AND

M.S. CIVIL & ENVIRONMENTAL ENG'G

EXECUTIVE EDUCATION CERTIFICATE

IN ENTREPRENEURSHIP & INNOVATION

Sep. 2019 - Jun. 2022 | Stanford, CA, USA

Knight-Hennessy Scholar (\$240k) (69/4,424)

UNIVERSITY OF THE PHILIPPINES

B.S. CIVIL ENGINEERING

Jun. 2013 - Jan. 2018 | Los Baños, PH

magna cum laude, college valedictorian (1/341)

completed one-semester early

TOKYO METROPOLITAN UNIVERSITY

EXCHANGE PROGRAM CERTIFICATE,

INTEGRATED RIVER ENGINEERING WORK

Oct. - Nov. 2017 | Tokyo, JP

SELECTED TALKS

Moody's Risk Management Solutions (RMS),
Global Earthquake Model (GEM) Conference,
RiskLayer, LandAware Innovations Group WG5

PEER-REVIEWED WORKS

Nature Communications, AAAI Social Impact,
ICML ML for Remote Sensing, Environmental
Research Letters, Progress in Disaster Science,
Sustainable Cities and Society (*in review*),
ISPRS Photo. & Remote Sensing (*in review*),
World Conf. Earthquake Engineering (WCEE),
Int'l Disaster Risk Conf. (IDRC)

SCIENTIFIC REVIEWER

AAAI AI for Social Impact, ICLR ML for Remote
Sensing, UNDRR-ISC Hazard Info Profiles,
Deep Learning Indaba, Cambridge J. of Sci. &
Policy, Stanford J. of Sci., Tech., & Society

AWARDS

2026 Digital GreenTalents Award
2026 EGU Outstanding Presentation
2025 Best Paper, Int'l Disaster Risk Conf.
2024 Helmholtz Researcher Grant
2023 AGU Outstanding Presentation
2018 BPI-DOST Science Award

SKILLS

MATLAB • QGIS • Google Earth Engine
Python • R • Stata • Tableau • $\text{\LaTeX}$

CERTIFICATION

Philippine Licensed Civil Engineer, 2018-2029

EXPERIENCE (8 1/2 years of research dev't, project mgm't, & risk consulting)

MIDDLESEX UNIVERSITY | BRITISH COUNCIL FELLOW IN AI & CLIMATE RESILIENCE

Oct. 2026 - Sep. 2027 | London, UK | 1 year

- Developing geospatial AI methods to integrate building vulnerability, exposure, and flood hazard data for spatiotemporal risk assessment.

UNIVERSITY OF CAMBRIDGE | DOCTORAL RESEARCHER, RISK & RESILIENCE

Oct. 2022 - Sep. 2026 | Cambridge, UK | 4 years

- Improved the spatiotemporal resolution of physical exposure models across 44+ Least Developed Countries by 5x finer using geospatial AI on time-series satellite imagery.
- Forged a university-industry partnership in the Philippines to deliver city-scale multi-hazard urban risk knowledge products.

GERMAN AEROSPACE CENTER (DLR) | GUEST SCIENTIST, DIGITAL GREEN TALENTS

Apr. - Jun. 2026 | Munich, Germany | 3 months

- Processed 28 GB of census data (543M+ rows) to produce the first nationwide maps of building typology at national, provincial, city, and barangay scale.

GERMAN AEROSPACE CENTER (DLR) | GUEST SCIENTIST, HELMHOLTZ AI

Jul. - Sep. 2024 | Munich, Germany | 3 months

- Designed a deep learning framework leveraging satellite imagery & census constraints to improve spatial disaggregation for building exposure and typology mapping.

EARTHQUAKES & MEGACITIES INITIATIVE (EMI) | URBAN RESILIENCE FELLOW

Mar. - Aug. 2022 | Quezon City, Philippines | 6 months

- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population under a scenario "Big One" M7.2 earthquake.

STANFORD UNIVERSITY, URBAN RESILIENCE INITIATIVE (SURI) | RESEARCHER

Dec. 2021 - May 2022 | Stanford, CA, USA | 6 months

- Conducted a probabilistic seismic risk assessment of 23 cities & municipalities in the Greater Metro Manila Area, Philippines, and benchmarked their disaster risk management capacities against the country's competitiveness index to evaluate policy priorities.

STANFORD LAND, BUILDINGS, & REAL ESTATE (LBRE) | SEISMIC RISK CONSULTANT

Mar. 2021 - Mar. 2022 | Stanford, CA, USA | 1 year

- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings under seismic scenarios up to magnitude 7.6, and developed a stakeholder-facing Tableau dashboard to support institutional decision-making.

FM GLOBAL, ENG'G & RESEARCH | STRUCTURES & NATURAL HAZARDS RESEARCHER

Jun. - Sep. 2021 | Norwood, MA, USA | 4 months

- Expanded the firm's library of earthquake design hazard maps by more than 246%, from 130 to 320 maps (190 new maps) across over 130 countries.

STANFORD UNIVERSITY, STRUCTURES AS SENSORS LAB | RESEARCHER

Jun. 2020 - Jun. 2021 | Stanford, CA, USA | 1 year

- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance post-earthquake USGS landslide, liquefaction, & building damage models by 15x.

ARUP | STRUCTURAL DESIGN ENGINEER

Jul. 2018 - Jul. 2019 | Metro Manila, Philippines | 1 year

- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41-50 story high-rise developments.

AYALA GROUP | PLANNING & CONTROL ENGINEERING INTERN

May. - Jul. 2017 | Bonifacio Global City, Philippines | 3 months

- Built a VBA-Excel system tracking deliverables across 100+ concurrent projects, improving project management and monitoring efficiency.

ENGINEERING CONGRESS | COMMITTEE HEAD

Jun. 2016 - Jun. 2017 | Los Baños, Philippines | 1 year

- Developed event invitation and registration strategies that increased participation fivefold (300 to 1,500 attendees) for major engineering seminars and quiz competitions.